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CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)

CHARUSAT Campus, Changa – 388 421.

M Sc (Biotechnology / Microbiology)
SEMESTER I EXAMINATION
BT / MI 701 : General Microbiology

Date: 26.04.2011
Time: 10.00 AM to 1.00 PM **Total Marks: 70**

Instructions to the candidates:

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
5. Use of cell phones is strictly prohibited.
6. Use of non-programmable calculators may be allowed if needed.

PART A **Total marks: 20**

Q1. Choose the correct option and put √ mark in front of it:

1. Heat sensitive media components are sterilized by
   (a) autoclaving (b) hot air oven (c) filtration (d) none of the above

2. The following are the examples of differential staining
   (a) Gram's & Acid fast (b) Gram's & Capsule
   (c) Gram's & Feulgen's (d) all of the above

3. The following was the first successful solid medium used for isolation of bacteria:
   (a) Agar (b) Potato (c) Gelatin (d) Meat

4. The first observation that bacteria-like organisms could be found in normal air was by
   (a) Anton Leeuwenhoek (b) Louis Pasteur
   (c) Robert Koch (d) Joseph Meister

5. The sequences that read the same on the complementary strand are called
   (a) complementary (b) palindromes (c) repetitive (d) none of the above

6. The increase in sequence dissimilarity after divergence may not be due to
   (a) parallel substitution (b) back substitution
   (c) convergent substitution (d) all of the above

7. The following techniques are useful in determining species, but not genera or families of an organism
   (a) % G C (b) Protein electrophoretic patterns
   (c) chemical composition (d) Serological techniques

8. Pseudogenes are studied to determine the pattern of spontaneous mutations under
   (a) adoptive pressure (b) selective neutrality
   (c) directed mutagenesis (d) all of the above

9. The organisms belonging to the following bacterial phylum use hydrogen for energy production and reduce oxygen
   (a) *Cyanobacteria* (b) *Chloroflexi* (c) *Thermotogae* (d) *Aquificae*

10. Genetic diversity of microbial community is assessed by measuring the following of the DNA from the entire community:
    (a) homogeneity (b) heterogeneity (c) G + C content (d) all of the above

11. The following is an example of microcosm:
    (a) Fermenter (b) Incubator (c) Winogradsky column (d) none of the above

12. The members of the Phylum *Planctomycetes* are unique as they:
    (a) Do not have peptidoglycan (b) Produce stalks
    (c) Divide by budding (d) All of the above

13. Clamp connection is found in the following class of fungi
    (a) Ascomycota (b) Deuteromycota (c) Basidiomycota (d) Zygomycota

14. Trisporic acid suppressed asexual reproduction in *Mucorales* ☐ True ☐ False

15. The following technique is used for the determination of microbial growth
    (a) cell forming units (b)microbial dry weight
    (c) spectrophotometry (d) all of the above

16. Succinyl CoA is not produced in cyanobacteria as they lack
    (a) Succinyl CoA synthatase (b)Succinic dehydrogenase
    (c)α–ketoglutarate dehydrogenase (d) citrate synthase

17. Blood agar is differential but not a selective medium. ☐ True ☐ False

18. In the following growth phase bacteria grows at optimum rate and population doubles rapidly
    (a) lag phase (b) log Phase (c) stationary phase (d) all of the above

19. The following relieves the insufficiency of nutrients, the accumulation of toxic substances, and excess cells in the culture.
    (a) turbidostat (b) batch culture (c) chemostat (d) all of the above

20. The turbidostat operates best at the following dilution rates
    (a) high (b) low (c) intermediate (d) all of the above

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Part B **Total Marks: 50**

Section I

Q.1.A. Write a short note on any **TWO** of the following: **(05)**
(i) Contributions of Robert Koch
(ii) Preservation of microorganism
(iii) Sterilization by chemical method

Q.1.B. Briefly write note on the following **(Any Two)** **(05)**
(i) Mechanism of gram's staining
(ii) Principle of Light microscope
(iii) Specimen preparation for electron microscopy

Q.2. A. Differentiate between [Any **Three**] **(06)**
(i) Classical Taxonomy and Chemical Taxonomy
(ii) Jukes and Cantor's model and Kimura's model
(iii) Neutral mutations and adoptive mutations
(iv) Slipped strand mispairing and unequal crossingover
(v) Transposition and Retroposition

Q.2. B. Describe briefly [**Any Two**] **(04)**
(i) Effects of transposition on the host genome.
(ii) Multigene family and Supragene
(iii) Serial endosymbiotic theory for evolution of various cell organelles

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Section II

Q.1. A. Explain briefly **(Any Three)** **(06)**

(i) Need of axenic culture to study microbial diversity in an ecosystem
(ii) Bacterial speciation
(iii) FISH
(iv) Evolutionary distance

Q.1.B. Justify **(Any Two)** **(04)**

(i) Species diversity is low in physically controlled ecosystem than that in Biologically controlled ecosystem.
(ii) Integrated approach is required for the study of microbial diversity
(iii) Presence of plasmid that is responsible for a phenotypic traits may lead to over-estimation of relatedness of one organism to another.

Q.2.A. Write note on the following **(Any Two)** **(06)**

(i) What is parasexuality? Briefly describe the event associated with it?
(ii) Role of secondary and tertiary hyphae in *Basidiomycota*
(iii) Economic importance of Rhodophyta

Q.2.B. Describe briefly: **(Any Two)** **(04)**

(i) The role of trisporic acid in sexual reproduction in *Zygomycota.*
(ii) The life cycle of chlamydomonas
(iii) Regulation of heterocyst formation

Q.3 A. Write note on the following **(Any Two)** **(06)**

(i) Methods used to obtain synchronous growth
(ii) Nutritional categories of microorganisms
(iii) Factors effecting microbial growth

Q.3 B. Differentiate between the following **(Any Two)** **(04)**

(i) selective and differential media
(ii) complex and simple media
(iii) log and lag phase of bacterial growth